Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/26/2022
PI Name:	Narayanan, Ranga Ph.D.		
Project Title:	A Novel Way to Measure Interfacial Tension Using the Electrostatic Levitation Furnace (ELF)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/22/2017	**End Date:**	05/21/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX17AL27G		
Performance Goal No.:			
Performance Goal Text:			
	A novel method using an Electrostatic Levitator (EL) is proposed to determine the surface tension of liquid/gas systems. This key thermophysical property is key to predicting optimum conditions for additive manufacturing, materials processing, and welding operations in space and on ground. The novel method for measurement of surface tension proposed involves the use of electrostatically forced Faraday instability. In this method, an oscillatory electrostatic forcing is applied to a liquid system, resulting in sudden pattern formation at the interface, i.e., an instability is caused. The "instability" is a result of resonance between the imposed forcing frequency and the natural frequency of the system. The critical amplitude and mode shapes are predictable from theory and compared with experiments. The measurement of surface tension directly aligns with the goals of the Thermophysical Property Measurement emphasis of the current NRA (NASA Research Announcement).		

Task Description:	Faraday experiments by this team using mechanical forcing have shown excellent agreement between theory and experimental results. However, in the case of mechanical forcing, the force of gravity overwhelms the effects of surface tension on the system stability and therefore it is almost impossible to determine this property with accuracy. Electrostatic forcing counters gravity, allowing the effects of surface tension to play a significant role in the ultimate determination of the system stability. Accordingly, experimentally mapped stability curves can be fit accurately with theoretical curves, allowing for the determination of the surface tension with great accuracy. The proposing team has successfully used electrostatic forcing in ground equipment to show that Faraday waves can be obtained as per theoretical predictions. The current state of the art uses an open container geometry to hold the liquid. However, a levitated sphere may also be used, and our proposal uses this containerless method. We therefore propose experiments with an important novel aspect. We propose to use ground-based facilities at JAXA (Japan Aerospace Exploration Agency) and NASA Marshall Space Flight Center (MSFC) to obtain tension for liquids using electrostatic Faraday forcing in using spherical drops by oscillating the drop over a large range of frequencies until the drop resonates with different modal patterns. The surface tension obtained from resonance at one modal patterns will be verified from deduced surface tension from another modal patterns. This will provide a self-consistent bench mark for determining surface tension. In summary, the research allows an alternate new method, viz., Faraday instability, for the determination of surface tension. Using electrostatic forcing instability will permit an independent confirmation of measurements obtained by the single-pulse oscillating-drop technique where decay rates are measured and related to surface tension and viscosity. The proposed method will also provide a slef consistent bench mark. Ground-based research using the electrostatic levitator available at the NASA MSFC and JAXA, and space experiments on ELF in the International Space Station (ISS) will provide accurate data. In addition to being an alternative method for surface tension measurement, the proposed technique also has versatility as a potential means in measuring surface tension for high viscosity materials and two-liquid interfacial tension on ground as well as in space. The proposing team has recently collaborated successfully with JAXA and has conducted experiments to study the feasibility of electrostatic Faraday instability. These results form the basis for the current proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Accurate surface tension values lead to transformative impact with space and Earth benefits such as a) Additive manufacturing of metals, where knowledge of surface tension is pivotal for predictive modeling to avoid rippling during solidification and possible wrinkling and shrinkage.. Human Exploration: Technology Area (TA) TA7.2.3.5: Welding, TA12.1.2: Computationally-Designed Materials, TA12.4.2.5: Additive Manufacturing. b) Space enabling operations. One example is joining operations such as soldering, where the surface plays an important role in the structure and stability of the weld. Accurate values for the surface tension will enhance the safety and effectiveness of this type of operation in space where the microstructures formed are different than those on Earth where gravity is present. c) Semiconductor crystal growth, where the interfacial tension between a liquid melt and liquid encapsulant plays a large role in determining conditions where convection is present.
Task Progress:	The main tasks during this reporting period were: a) finalize the flight samples to be tested on the ELF (Electrostatic Levitation Furnace), b) finalize the experimental protocols for sample testing. To accomplish these tasks, several experiments were conducted on site at the NASA Marshall Space Flight Center (MSFC), using Zirconium, Tin, and Indium. The basic principles of excitation of different resonant modes and correlation of resonant frequencies to surface tension were done. This assured the correctness of the experimental protocols to be used during flight testing.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Abstracts for Journals and Proceedings	Livesay M, Brosius N, SanSoucie M, Matsumoto S, Ishikawa T, Narayanan R. "Self-benchmarked surface tension measurement using multi-mode resonance of levitated molten droplets." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021
Abstracts for Journals and Proceedings	Dinesh B, Brosius N, Corbin T, Narayanan R. "The role of a deep wavy wall on the natural frequencies of a fluid interface: Theory and experiments." 74th Annual Meeting of the APS Division of Fluid Dynamics, Phoenix, AZ, November 21–23, 2021. Bulletin of the American Physical Society. 2021 Nov;66(17):Abstract: T10.00001. https://meetings.aps.org/Meeting/DFD21/Session/T10.1 , Nov-2021
Articles in Peer-reviewed Journals	Dinesh B, Narayanan R. "Nature of branching in electrohydrodynamic instability." Phys. Rev. Fluids. 2021 May 20;6(5):054001. https://doi.org/10.1103/PhysRevFluids.6.054001 , May-2021
Articles in Peer-reviewed Journals	Pillai D, Sahu K, Narayanan R. "Electrowetting of a leaky dielectric droplet under a time-periodic electric field." Phys. Rev. Fluids. 2021 Jul 2;6(7)073701. https://doi.org/10.1103/PhysRevFluids.6.073701 , Jul-2021
Articles in Peer-reviewed Journals	Dinesh B, Corbin T, Narayanan R. "Thin-film Rayleigh-Taylor instability in the presence of a deep periodic corrugated wall." J. Fluid Mechanics. 2022 Jan 25;931(R5). https://doi.org/10.1017/jfm.2021.960 , Jan-2022
Dissertations and Theses	Brosius N. "Resonance in microgravity transport processes: Measurement of surface tension and heat transfer enhancement." Dissertation, University of Florida, August 2021. , Aug-2021